

50-amaliy mashg'ulot

Na'muna uchun misollar yechimi

1-misol. Ushbu

$$u = x^y$$

funksiyaning differensial topilsin.

◀ Ravshanki,

$$\frac{\partial u}{\partial x} = yx^{y-1}, \quad \frac{\partial u}{\partial y} = x^y \ln x.$$

Unda (5) formulaga ko'ra

$$du = yx^{y-1}dx + x^y \ln x dy$$

bo'ladi. ►

2-misol. Tomonlari $x=6m$ va $y=8m$ bo'lgan to'g'ri to'rtburchak berilgan. Agar bu to'g'ri to'rtburchakning x tomonini 5 sm. ga oshirilsa, u tomonini 10 sm. ga kamaytirilsa, to'rtburchakning diagonalini qanchaga o'zgaradi?

◀ Agar berilgan to'g'ri to'rtburchakning diagonalini u desak, unda

$$u = \sqrt{x^2 + y^2}$$

bo'ladi. Endi

$$\Delta u(x_0, y_0) \approx \frac{x_0}{\sqrt{x_0^2 + y_0^2}} \Delta x + \frac{y_0}{\sqrt{x_0^2 + y_0^2}} \Delta y$$

bo'lishini e'tiborga olib, topamiz:

$$\Delta u(x_0, y_0) \approx \frac{x_0}{\sqrt{x_0^2 + y_0^2}} \cdot \Delta x + \frac{y_0}{\sqrt{x_0^2 + y_0^2}} \cdot \Delta y = \frac{x_0 \cdot \Delta x + y_0 \cdot \Delta y}{\sqrt{x_0^2 + y_0^2}}$$

Bu munosabatda

$$x_0 = 6m, \Delta x = 0,05m, y_0 = 8m, \Delta y = -0,10m$$

deyilsa, unda

$$\Delta u \approx \frac{6 \cdot 0,06 + 8 \cdot (-0,10)}{\sqrt{36 + 64}} m = -0,05m$$

bo'lishi kelib chiqadi.

Demak, to'g'ri to'rtburchakning diagonali taxminan 5 sm. ga kamayar ekan. ►

3-misol. Ushbu

$$\alpha = 1,07^{3,97}$$

Miqdor taqribiy xisoblansin.

◀ Ravshanki, $\alpha = 1,07^{3,97}$ son ushbu

$$f(x, y) = x^y$$

Funksiyaning $x = 1,07$, $y = 3,97$ dagi xususiy qiymatidan iborat. Bu funksiya uchun $f(1, 4) = 1$ bo'lishini e'tiborga olib

$$x_0 = 1, \quad y_0 = 4$$

deb olamiz. Unda

$$\Delta x = x - x_0 = 1,07 - 1 = 0,07,$$

$$\Delta y = y - y_0 = 3,97 - 4 = -0,03$$

bo'ladi. Ushbu

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0, y_0) + f'_x(x_0, y_0) \cdot \Delta x + f'_y(x_0, y_0) \cdot \Delta y$$

formuladan foydalanib topamiz:

$$f(x_0 + \Delta x, y_0 + \Delta y) = f(1 + 0,07, 4 - 0,03) = f(1,07, 3,97) = 1,7^{3,97}$$

$$f(x_0, y_0) = f(1, 4) = 1^4 = 1$$

$$f'_x(x_0, y_0) = \frac{\partial}{\partial x}(x^y) \Big|_{x_0, y_0} = y \cdot x^{y-1} \Big|_{x=1, y=4} = 4 \cdot 1 = 4$$

$$f'_y(x_0, y_0) = \frac{\partial}{\partial y}(x^y) \Big|_{x_0, y_0} = x^y \ln x \Big|_{x=1, y=4} = 1^4 \cdot \ln 1 = 0$$

$$1,7^{3,97} \approx 1 + 4 \cdot 0,07 + 0 \cdot (-0,03) = 1 + 0,28 = 1,28$$

Demak,

$$\alpha = 1,7^{3,97} \approx 1,28. \blacktriangleright$$

Misollar

Quyidagi funksiyalarning differensiallari topilsin.

1. $z = x^2 y - xy^2 + 3$

2. $z = (x^2 + y^2)^3$

3. $z = \sin^2 x + \cos^2 y$

4. $z = \arctg(xy)$

5. $z = e^{12x+5y}$

6. $z = (\sin x)^{\cos y}$

7. $z = \ln \left(1 + \frac{x}{y} \right)$

8. $z = \frac{x}{y} e^{xy}$

9. $x = \arctg \sqrt{xy}$

10. Agar $f(x, y) = \frac{x}{y^2}$ bo'lsa, $df(1,1)$ topilsin.

11. Ushbu $f(x, y) = e^{xy}$ funksiya to'liq differensialini $x=1, y=2, dx=-0,1, dy=0,1$ bo'lgandagi qiymati topilsin.

12. Ushbu $f(x, y) = \frac{x}{x-y}$ funksiya to'liq differensialini $x=2, y=1, dx=-\frac{1}{3}, dy=\frac{1}{2}$ bo'lgandagi qiymati topilsin.

Quyidagi miqdorlar taqribiy hisoblansin.

13. 1) $1,08^{3,96}$,

2) $1,94e^{0,12}$

14. 1) $\sin 1,59 \cdot \lg 3,09$,

2) $2,68e^{\sin 0,05}$

